Randomized and low-rank approximation

Optimal matvec query complexity of HODLR approximation

Difficulty: challenging **Importance:** interesting to the community

Status: Partially resolved

Rating rationale: Challenging because matching adaptive-query lower and upper bounds must cover depth and accuracy jointly; community impact is the cost of hierarchical matrix compression.

Area: information complexity of hierarchical matrix approximation

Partial resolution — 13 September 2026

Sidney Holden (Center for Computational Biology, Flatiron Institute, Simons Foundation) proves the following bounds in Theorem 1.1 of the continuation manuscript, for all original parameters and universal constants $c, C > 0$:

$$c \min\{n, kL/\varepsilon + k/\varepsilon^2\} \leq q_*(n, k, \varepsilon) \leq C \min\{n, kL^2/\varepsilon + kL/\varepsilon^2\}.$$

The upper bound is nonadaptive and returns a proper HODLR approximation with the original fixed-input success guarantee. Corollary 1.2 gives $q_* = \Theta(n)$ when $\varepsilon \leq \sqrt{k/n}$. A separate independent Codex AI-agent informal audit passed these partial results, including the adaptive two-sided lower bounds. Submission, verified affiliation, sources and reproduction limits.

Still open: determine the general joint rate up to universal constants. The uncapped upper expression is L times the lower expression, so these results do not solve the full original target. The problem remains in the open count. The available five-case assembly smoke suite passed; the historical continuation verification suite is incomplete in the supplied archive. No Lean verification, external human peer review or historical novelty claim is asserted.

Context and notation

Use exact real arithmetic, with comparisons, standard Gaussian sampling, and exact SVDs available as primitives; charge an SVD of an $a \times b$ matrix $O(ab \min(a, b))$ operations. This states the idealized arithmetic model used here, rather than a finite precision or bit complexity claim. A matrix–vector query returns either Av or $A^{\mathsf{T}}v$ for one chosen real vector v; both types count toward the total. Randomized guarantees are for every fixed input, with probability or expectation over the algorithm's randomness.

Problem statement

Let $n = 2^L k$ with integers $k \geq 1, L \geq 2$, and $0 < \varepsilon < 1/2$. Define $\mathcal{H}_{n,k}$ recursively: a matrix of order at most k is unrestricted; at a larger node, the two equally sized diagonal blocks must satisfy the same definition, and each off-diagonal block must have rank at most k.

Let $q_*(n, k, \varepsilon)$ be the smallest worst-case number of queries with which a randomized adaptive algorithm, given only oracle access to an arbitrary $A \in \mathbb{R}^{n\times n}$, returns $B \in \mathcal{H}_{n,k}$ such that

$$\Pr\left[\|A - B\|_F \leq (1 + \varepsilon) \min_{H\in\mathcal{H}_{n,k}} \|A - H\|_F\right] \geq 0.99$$

for every A. Here only oracle calls are charged: arbitrary measurable computations between queries are allowed. This is an information complexity question.

Question

Determine $q_*(n, k, \varepsilon)$ up to universal multiplicative constants, including its joint dependence on k, L, ε.

Reference

Tyler Chen, Feyza Duman Keles, Diana Halikias, Cameron Musco, Christopher Musco, and David Persson, *Near-optimal hierarchical matrix approximation from matrix-vector products*, SODA 2025, 2656–2692, Definition 1.1, Problem 1.1, Theorems 1.1–1.2, end of §6, and §8.

The known upper bound is $O(\min\{n, kL^4/\varepsilon^3\})$. The source proves $\Omega(kL + k/\varepsilon)$ in its regime $n \geq c_0 k/\varepsilon$ for a sufficiently large absolute c_0; this restriction matters near the full-recovery threshold.

Status evidence

The latest arXiv version is v2, October 24, 2024. Searches for "HODLR query complexity 2026" and "HODLR approximation 2026 lower bound" found no matching improvement. Musco's February 2026 ICERM slides, numbered slide 14, retain the stated upper bound. No later explicit open-status confirmation was located.

Audit — 2026-09-10

Rechecked Chen et al., Theorems 1.1–1.2 and §8. The gap in logarithmic and accuracy factors remains. HODLR query-complexity and later hierarchical-approximation searches found no matching characterization, including the near-full-recovery regime.